OBD2 Viewer — Quick Start User Guide

Your Complete Vehicle Diagnostics & Trip Computer

📱 What is OBD2 Viewer?

OBD2 Viewer transforms your Android phone into a powerful vehicle diagnostics tool and trip computer. Connect to your car's OBD-II port via Bluetooth, view live engine data, track trips with GPS, and analyze your driving with customizable dashboards.

🚀 Quick Start (5 Steps)

1. **Hardware Setup**

- Plug your **ELM327 Bluetooth adapter** into your vehicle's OBD-II port (usually under the dashboard near the steering wheel)
- Turn on your vehicle's ignition (engine can be off for initial setup)
- **Pair the adapter** with your Android device via Settings → Bluetooth (typically shows as "OBDII" or "ELM327")

2. **First Launch**

- Open OBD2 Viewer
- Grant **Bluetooth** and **Location** permissions when prompted
- Enable Bluetooth if prompted

3. **Connect to Your Vehicle**

- Tap the **Connect** tab
- Find your paired OBD-II adapter in the device list
- Tap the device name to connect
- Wait for "Connected" status (takes 5-10 seconds)

4. **Create Your Vehicle Profile** (Optional but Recommended)

- Go to **Settings** tab
- Tap **"+ Add Profile"**
- Enter: Vehicle name, fuel type, tank capacity, fuel price
- Advanced: Engine power (BHP), vehicle mass (kg) for power calculations
- Tap **Save**

5. Start Your First Trip

- Go to **Trip** tab
- Tap **"Start Trip"** button
- Drive normally — the app tracks everything automatically
- Tap **"Stop Trip"** when done

📊 Main Features

🎯 Trip Computer

- **Real-time metrics**: Distance, duration, fuel used, average speed
- **Fuel efficiency**: Instant and trip average (L/100km or km/L)
- **Drive mode analysis**: % time in city/highway/idle
- **GPS tracking**: Accurate distance and speed with satellite positioning
- **Trip logging**: Save detailed JSON logs for analysis

📊 Live Dashboard

- **Customizable layouts**: Create unlimited dashboard configurations
- **6 widget types**: Dial gauges, 7-segment displays, bar gauges, numeric displays, temperature arcs
- **50+ metrics**: RPM, speed, temperatures, fuel, power, acceleration, and more
- **Color schemes**: Dark, Neon Red, Green LCD themes
- **Warning zones**: Set thresholds for visual alerts

🔧 Diagnostics

- **21 OBD-II parameters**: Engine load, coolant temp, throttle, MAF, fuel trim, O2 sensors, and more
- **Auto-discovery**: Only displays parameters your vehicle supports
- **Details view**: Complete table of all live sensor readings

📍 GPS Integration

- **Accurate tracking**: Uses Google Play Services location
- **Altitude correction**: MSL altitude with geoid correction
- **Satellite count**: Shows GPS signal quality
- **Speed validation**: Cross-checks OBD speed vs GPS speed

📱 Accelerometer Analysis (Optional)

- **Road quality tracking**: Detects bumps and potholes
- **G-force measurement**: Forward/lateral/vertical acceleration

- **Lean angle**: Vehicle tilt during cornering
- **Brake/acceleration peaks**: Maximum forces recorded

🎮 Using the App

Navigation

Swipe between 5 main screens or use the bottom navigation bar: 1. **Trip** — Start/stop trips, view live stats 2. **Connect** — Bluetooth device connection 3. **Dashboards** — Customizable gauge layouts 4. **Details** — Full OBD-II parameter table 5. **Settings** — Vehicle profiles and preferences

Recording a Trip

Start: 1. Ensure OBD-II adapter is connected (green indicator on Trip screen) 2. Tap **"Start Trip"** 3. Drive normally — all data is recorded automatically 4. App runs in background with persistent notification

Pause/Resume: - Tap **"Pause"** to temporarily stop recording (e.g., at a gas station) - Tap **"Resume"** to continue the same trip - Paused time is excluded from trip calculations

Stop: - Tap **"Stop Trip"** to end recording - Trip log is automatically saved (if logging enabled in Settings) - View trip summary before closing

Creating Custom Dashboards

1. Go to **Dashboards** tab → Tap **menu (⋮)** → **"New Layout"**
2. Choose a name and color scheme
3. Tap **"+ Add Widget"**
4. Select:
    - **Metric** (e.g., RPM, Speed, Coolant Temp)
    - **Widget type** (Dial, 7-Segment, Bar, etc.)
    - **Range** (min/max values, auto-populated)
    - **Warning threshold** (optional)
5. **Drag** widget to position, **resize** using corner handles
6. Tap **Save** when done
7. Switch between layouts via the layout list

Managing Vehicle Profiles

Why use profiles? - Different vehicles have different fuel types, tank sizes, and engine specs - Profiles enable accurate fuel cost and power calculations - Set custom OBD polling rates per vehicle

Create a profile: 1. Settings → **"+ Add Profile"** 2. Fill in basic info (name, fuel type, tank capacity, fuel price) 3. Optional: Engine power (BHP) and vehicle mass (kg) for power metrics 4. Tap **Save**

Switch profiles: - Tap any profile in the list to make it active - Active profile shows a blue badge

---

⚙️ Settings Explained

Setting	What it does
OBD Connection	Toggle between real OBD adapter and mock mode (for testing)
Auto-connect	Automatically connect to last used device when opening Connect screen
Enable Logging	Save trip data to JSON files in selected folder
Auto-share Log	Automatically open share dialog after trip ends
Log Folder	Choose where trip logs are saved (default: Downloads)
Accelerometer	Enable road quality and G-force tracking (uses more battery)
Bluetooth Logging	Save raw Bluetooth communication for debugging

---

💡 Tips & Best Practices

For Best Results:

✅ **Keep GPS enabled** — Required for accurate distance and speed
✅ **Create a vehicle profile** — Enables fuel cost and power calculations
✅ **Start trip after engine starts** — Ensures all sensors are active
✅ **Keep phone charged** — GPS + Bluetooth drain battery faster
✅ **Use a phone mount** — Safely view dashboard while driving

Troubleshooting:

Can't connect to adapter? - Verify adapter is paired in Android Bluetooth settings first - Ensure vehicle ignition is ON - Try unplugging and re-plugging the adapter - Check adapter LED is blinking (indicates power)

No GPS signal? - Ensure Location permission is granted - Move away from buildings/tunnels for initial GPS lock - Wait 30-60 seconds for satellite acquisition

Missing OBD parameters? - Your vehicle may not support all 21 parameters - Only ECU-supported parameters are displayed - Older vehicles support fewer parameters

Dashboard widgets not updating? - Ensure OBD adapter is connected (check Connect screen) - Verify the metric is supported by your vehicle - Check Details screen to see raw values

Battery draining fast? - Disable accelerometer if not needed (Settings) - Reduce screen brightness - Close other GPS apps running in background

---

Trip Logs

Format: JSON files with complete trip data
Filename: `<ProfileName>_obdlog_<YYYY-MM-DD_HHmmss>.json`
Location: Chosen log folder (or Downloads by default)
Contains: All OBD-II readings, GPS coordinates, accelerometer data, trip statistics

Viewing logs: - Use any JSON viewer or text editor - Import into spreadsheet software for analysis - Share via email, cloud storage, or USB

Permissions Required

Permission	Why it's needed
Bluetooth	Connect to OBD-II adapter
Location	GPS tracking for distance/speed + required for Bluetooth scanning on Android 12+
Foreground Service	Keep trip recording active in background

Privacy: All data stays on your device. No data is sent to external servers.

System Requirements

- **Android 8.0** or higher
- **Bluetooth Classic** support (all modern phones)
- **GPS** for trip tracking
- **Accelerometer** (optional, for road quality analysis)

Compatible Adapters: - Any **ELM327 Bluetooth** adapter (Classic BT, not BLE) - Recommended: BAFX, OBDLink, Veepeak brands

---

🆘 Support & Resources

App Version: Check Settings screen for current build number
Documentation: See `README.md` and `ARCHITECTURE.md` in app repository
Versioning: Build numbers auto-increment — use for bug reports

Common Questions:

Q: Can I use this with multiple vehicles?
A: Yes! Create a separate profile for each vehicle and switch between them.

Q: Does this work with diesel engines?
A: Yes, select "Diesel" as fuel type in your vehicle profile.

Q: Will this drain my car battery?
A: No. The OBD-II port is powered only when ignition is on. Always unplug the adapter when not in use for extended periods.

Q: Can I export trip data?
A: Yes, enable logging in Settings. Trip logs are saved as JSON files you can share or analyze.

Q: Why are some metrics showing "—" or blank?
A: Your vehicle's ECU doesn't support those parameters. Only supported sensors are polled.

🎯 Quick Reference Card

Action	How to do it
Connect to car	Connect tab → Tap device name
Start trip	Trip tab → Start Trip button
Pause trip	Trip tab → Pause button
Stop trip	Trip tab → Stop Trip button
View live data	Details tab (full table) or Dashboards tab (gauges)
Create dashboard	Dashboards → ⋮ menu → New Layout
Add vehicle	Settings → + Add Profile
Switch vehicle	Settings → Tap profile name
Enable logging	Settings → Enable Logging toggle
Choose log folder	Settings → Change Log Folder

Drive safe and enjoy your OBD2 Viewer experience! 🚗 💨